

Vascular Access for Hemodialysis

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Functional vascular access is needed for all extracorporeal dialytic therapies and is the lifeline for patients treated with chronic hemodialysis (HD). The three main forms of vascular access are the arteriovenous fistula (AVF), arteriovenous graft (AVG), and central venous catheter (CVC). Arteriovenous (AV) access refers to either an arteriovenous fistula or graft. The ideal AV access should have a suitable superficial conduit (autogenous or synthetic) that allows for two-needle cannulation that delivers blood flow for effective dialysis (Q_b), with intraaccess flows (Q_a) usually in the range of 400 to 1500 mL/min. A well-planned vascular access should be functional when needed for dialysis, have good primary patency, have few complications, and align with the patient's end-stage kidney disease (ESKD) Life-Plan.¹ Previously, it had been taught that the ideal access should be an AVF placed peripherally at the wrist. However, because upper arm and lower limb access sites or CVCs are increasingly used due to the patient's limiting anatomy or circumstances (e.g., frail elderly with multiple comorbidities, exhaustion of access sites, future transplant), a more comprehensive approach to vascular access choice was developed that considers the patient's ESKD Life-Plan.

ESKD LIFE-PLAN

The ESKD Life-Plan is an individualized map of kidney replacement modalities (hemodialysis [HD], peritoneal dialysis, transplantation) required during the patient's life with chronic kidney disease (CKD). All patients with progressive CKD and/or with estimated glomerular filtration rate (eGFR) of 15 to 20 mL/min/1.73 m² or already on kidney replacement therapy should have a Life-Plan. The Life-Plan is regularly reviewed and revised after considering the patient's unique medical and life circumstances and their informed preferences. The appropriate dialysis access is then chosen to align with and support the patient's ESKD Life-Plan; if done correctly, each patient would then have an individualized PLAN: *patient life-plan* and their *access needs*.¹ Within "access needs," there are three main components: (1) insertion/creation plan, (2) contingency plan (a plan to manage complications), and (3) succession plan. Concurrently, there must always be a vessel preservation plan to ensure viability for future access as required. Therefore, each vascular access must include four plans: vessel preservation plan, insertion/creation plan, contingency plan, and succession plan.¹

Vascular access should be created in a timely manner by a surgeon or interventionalist experienced in vascular access creation to avoid unnecessary CVC use and its associated complications.¹ However, vascular access complications are almost inevitable and may require management by endovascular procedures or surgical revision. A multidisciplinary approach to access creation and maintenance (involving nephrologists, interventional radiologists, access surgeons, and dialysis nurses) helps optimize HD access-related outcomes and minimize costs.

EVALUATION OF THE PATIENT FOR VASCULAR ACCESS

CKD patients should be considered for vascular access if they have progressive decline in kidney function and referred when eGFR is 15 to 20 mL/min/1.73 m². Earlier referral should occur in patients with unstable and/or rapid rates of GFR decline (e.g., >10 mL/min/1.73 m²/yr).¹ Patients with a failing kidney transplant or who are not responding to peritoneal dialysis should be assessed for vascular access in a timely manner to avoid CVC use when transitioning to HD. If an AV access is deemed appropriate based on the ESKD Life-Plan, the type and site of the vascular access will be based on the following:

- *Physical examination*, with careful inspection and palpation of arterial pulses and venous vasculature. Particular attention is paid to the venous filling capacity, with use of a blood pressure cuff and variable pressures, and to the presence of venous collaterals and swelling. The nondominant arm is not necessarily the preferred side, and the decision should be based on the quality of the vessels and future access needs.
- *Vascular mapping* is usually done with Duplex ultrasound and provides information about the venous vasculature (particularly in obese patients and in the upper arm) and the diameter of the brachial, radial, and ulnar arteries; detects vascular calcifications; and reveals the blood flow volume in the brachial artery. The resistance index (a measure of arterial compliance) can be calculated from the differences between the high-resistance triphasic Doppler signal with clenched fist and the low-resistance biphasic waveform after the fist is released. A preoperative resistance index of 0.7 or higher in the feeding artery indicates insufficient arterial compliance (often associated with arterial calcification), warning of lower success with AVF creation. Guidelines support vessel mapping in most patients (Table 96.1).
- Additional angiography and/or phlebography is needed only in very difficult cases or in patients with previous ipsilateral CVCs to rule out central vein stenosis or obstruction; the use of radiocontrast media should be minimized in patients with residual kidney function.

Preservation of vessels is crucial for the success of vascular access. Patients should be instructed to protect their veins, restricting blood sampling and intravenous cannulas to the dorsum of the hand whenever possible. Physicians that may access radial arteries for interventional procedures (e.g., cardiac investigations) should be alerted to consider alternate access for these procedures.

VASCULAR ACCESS TYPES

Autogenous Access

Distal Forearm Fistulas

Radiocephalic arteriovenous fistula. A well-functioning distal radiocephalic AVF in the nondominant arm is the ideal permanent

TABLE 96.1 Risk Factors for Which Vessel Mapping May Be Beneficial

Clinical Problem	Risk Factors
Fistula failure	Elderly age; female sex; comorbidities (e.g., peripheral vascular disease; coronary artery disease)
Peripheral vessel damage	PICC insertion; iatrogenic vein damage; radial artery harvesting for coronary artery bypass grafting
Central venous stenosis	Multiple CVC; prolonged CVC duration; cardiac implantable electronic device; PICC; surgery or trauma to neck, chest or upper extremity
Limitations to physical examination	Severe obesity; suboptimal conditions (dehydration or vasoconstriction); poor skin integrity

CVC, Central venous catheter; PICC, peripheral inserted central catheter.

access for HD. This usually gives adequate blood flow and a long length of superficial vein for needling. It also leaves proximal sites for further procedures in the event of failure. Although this distal radiocephalic AVF is the aim for incident patients, its creation and success may be compromised if the cephalic and antecubital fossa veins are damaged because of thrombophlebitis and/or thrombosis from previous intravenous cannulas or venipunctures.

A radiocephalic AVF is usually created at the wrist but can be created more proximally in the forearm if distal vessels are inadequate (Fig. 96.1). On occasion, three or four radiocephalic AVFs can be created at progressively more proximal sites in the forearm before a brachiocephalic AVF is created. The radiocephalic AVF at the wrist was initially described as a side-to-side anastomosis, but an end-to-side configuration is preferred to reduce the risk of venous hypertension in the radial aspect of the hand.

The patency of radiocephalic fistulas varies from center to center and from surgeon to surgeon. Reported primary failure rates vary from 5% to 60%, and 1-year patency rates vary from 52% to 71% (Table 96.2),²⁻⁸ primarily due to early thrombosis and AVF nonmaturation. Risk factors for early failure include diminished radial artery or cephalic vein size or quality, female sex, older age, greater and complex comorbidity, and center effect.^{9,10} Patients deemed at high risk for distal radiocephalic fistula primary failure should be considered for a proximal elbow or upper arm AVF or forearm graft, depending on the patient's ESKD Life-Plan.

Proximal Forearm and Upper Arm Fistulas

Although a radiocephalic AVF is preferable, the first created vascular access is increasingly an upper arm AVF with use of an autogenous superficially located arm vein,¹¹ especially in patients with comorbidities such as diabetes mellitus, coronary heart disease, and peripheral vascular disease.^{12,13}

The upper limb is preferred to the lower limb for vascular access because of the ease of cannulation, comfort for the patient, and considerably lower incidence of complications. Similarly, autogenous conduits are preferable to the use of prosthetic grafts because of better patency and lower risk of infection.

Forearm cephalic and basilic vein transposition. Vein transposition or elevation increases the possibilities for creating a forearm fistula. The cephalic vein is preferred, but if it is unsuitable, the more deeply located basilic vein can be transposed from the ulnar to the

Standard Radiocephalic AV Fistula at the Wrist

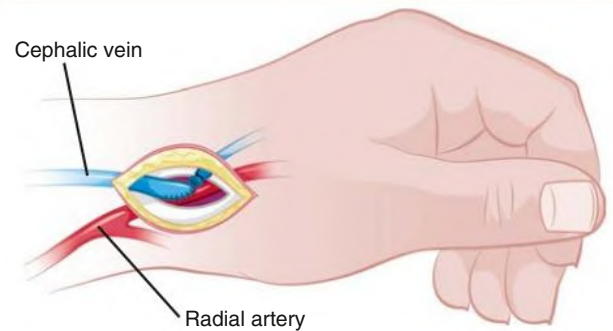


Fig. 96.1 Standard Radiocephalic Arteriovenous (AV) Fistula at the Wrist. Anastomosis of end of vein to side of artery.

radial side along a straight subcutaneous course from the elbow to the radial artery. Alternatively, an anastomosis from the basilic vein to the ulnar artery can be performed with additional volar transposition to facilitate needling for dialysis.

Different surgical techniques (with or without vein transposition) have been advocated according to the forearm artery and vein location, with the possibility of achieving 91% fistula maturation¹⁴; for example, in one study 15% of participants had a straightforward AVF, 33% required vein transposition from dorsal to volar, and 52% required superficial forearm vein transposition before arterial anastomosis. Primary patency rates were 84% at 1 year and 69% at 2 years. Cannulation of radiocephalic and/or brachiocephalic AVFs may be difficult, particularly in obese patients. Preplanned vein elevation or transposition techniques in such patients can be effective and should be considered; the primary failure rate has been reported to be 15%, with a 1-year patency rate of 84%. Alternatively, surgical lipectomy or liposuction can be performed to facilitate cannulation of deeply located AVF.¹⁵

Elbow and upper arm cephalic vein arteriovenous fistula. The brachiocephalic and antecubital configurations can be used for AV anastomoses in the elbow region. If possible, the anastomosis can be created between the cephalic, cubital, or perforating vein and the brachial artery, typically 2 cm distal to the elbow, which optimizes opportunities for cannulation along the cephalic vein in the upper arm (Fig. 96.2). The outcome of a brachiocephalic AVF is usually good, with high primary function and good long-term patency; studies show a 10% early failure rate caused by nonmaturation and an 80% 1-year patency rate.^{16,17} Two-year primary, assisted primary, and cumulative patency rates were 40%, 59%, and 67%, respectively. Primary patency means that the access is functioning without any intervention; assisted primary patency means that the access is functioning but required a preemptive intervention for flow decline. Cumulative or secondary patency refers to accesses that are functioning after intervention for thrombosis. The term *cumulative patency* is preferred to reduce confusion and improve standardization for communication about vascular access.^{1,18} The early failure and 1-year patency rates of brachiocephalic AVFs are shown in Table 96.3.^{16,17,19-21} Newer endovascular options for creation of antecubital AVF, typically between the perforating vein and ulnar or radial arteries, have recently been described. Early data are promising, demonstrating both technical success and 6-month patencies each greater than 88%.²²⁻²⁴

Upper arm basilic vein arteriovenous fistula. The upper arm basilic vein is usually inaccessible or painful for dialysis cannulation because of its medial and deep position. Therefore, the basilic vein needs to be superficialized and transposed to an anterolateral position. In the

TABLE 96.2 Early Failure and 1-Year Patency Rates of Radiocephalic Arteriovenous Fistulas

Study	Number of Fistulas	Early Failure (%)	1-Year Patency (%)
Ravani et al., 2002	197	5	71
Rooijens et al., 2005	86	41	52
Korten et al., 2007	148	11	57
Biuckians et al., 2008	80	37	63
Al-Jaishi et al., 2014	890	28	55
Goh et al., 2016	204	12	66
Wilmink et al., 2016	689	26	70

Options for the Creation of Elbow AV Fistulas

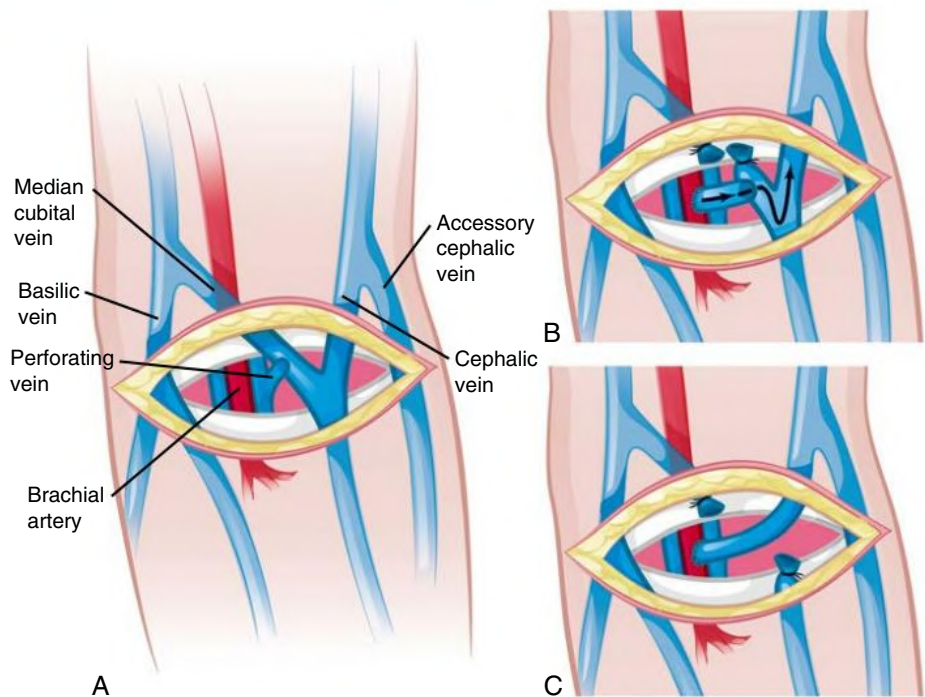


Fig. 96.2 Options for the Creation of Elbow Arteriovenous (AV) Fistulas. (A) Anatomy of arterial and venous vessels in the elbow. (B) Brachio-perforating vein AV fistula with ligation of proximal cubital vein. (C) Brachiocephalic AV fistula.

TABLE 96.3 Early Failure and 1-Year Patency Rates of Brachiocephalic Arteriovenous Fistulas

Study	Number of Fistulas	Early Failure (%)	1-Year Patency (%)
Zeebregts et al., 2005	100	11	79
Lok et al., 2005	186	9	78
Woo et al., 2007	71	12	66
Koksoy et al., 2009	50	10	87
Ayez et al., 2012	87	8	83

two-step approach to creating a brachio basilic AVF, the anastomosis is first constructed, followed by a second operation, usually 6 weeks later, whereby the arterialized vein is mobilized into a subcutaneous position to enable needling (Fig. 96.3). Alternatively, in the one-step approach to creating the brachio basilic AVF, the vein is elevated and transposed into position at the time of anastomosis creation. Primary failure rates are 2% to 26%, with 1-year cumulative patencies varying from 69% to 86%.²⁵

Brachio basilic AVF have several advantages over forearm or upper arm AV grafts, such as less infection and thrombosis. A meta-analysis found that brachio basilic AVFs have superior primary and cumulative patency compared with AV grafts²⁶ and should be preferentially considered in difficult access cases and when suitable according to the PLAN.¹

Lower limb vascular access. The main indications for lower limb vascular access are bilateral central venous or caval obstruction,

Transposed Brachiobasilic AV Fistula

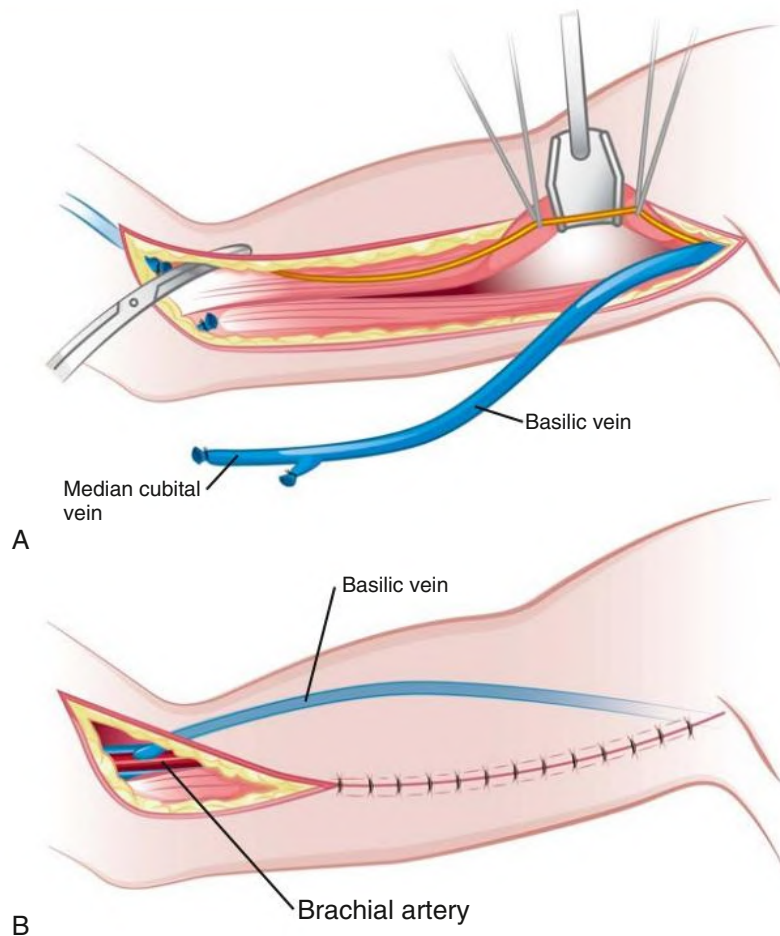


Fig. 96.3 Transposed Brachiobasilic Arteriovenous (AV) Fistula. (A) Dissection of the basilic vein. (B) Anterolateral transposition and brachial artery anastomosis.

which endangers the outflow of upper limb AVFs. Saphenous and superficial femoral vein transposition is a primary option for thigh AVFs, although this carries a relatively high risk for distal ischemia. If clinical evaluation indicates incipient ischemia, primary flow reduction by tapering of the anastomosis is indicated to prevent complications. Synthetic AV graft implantation in the thigh bears a high risk for infection and septicemia.²⁷

NONAUTOGENOUS SYNTHETIC VASCULAR ACCESS

Arteriovenous Grafts

An AV access (referring to AVF or AV graft) is preferable to a CVC, whenever feasible. When autogenous AVF creation is impossible, an AV graft should be considered. Xenografts such as the ovine (sheep) graft are an alternative access conduit with acceptable patency and low infection rates. However, the most frequently used synthetic grafts are made of expanded polytetrafluoroethylene (ePTFE) and, less commonly, polyurethane. These prosthetic grafts can be implanted in a wide variety of locations and configurations in the upper and lower limbs (Fig. 96.4). At present, early cannulation of AV grafts (within 24 hours of surgery) is feasible because of newer graft compositions. In randomized controlled trials (RCTs), they have been shown to be catheter sparing and have equivalent patencies to standard AV grafts.^{28,29}

The primary patency rates of AV grafts vary from 60% to 80% at 1 year and from 30% to 40% at 2 years of follow-up. Secondary patency ranges from 70% to 90% and from 50% to 70% at 1 and 2 years, respectively.³⁰⁻³³ The new KDOQI Vascular Access Guidelines 2019 have helpful algorithms for the appropriate use of early cannulation and standard AV grafts.¹

CENTRAL VENOUS CATHETER ACCESS

CVCs are widely used to initiate HD despite recommendations for AV access use.^{1,34,35} Data from the Dialysis Outcomes and Practice Patterns Study (DOPPS)³⁶ indicate that 15% of HD patients in the United States are dialyzed with catheters; in other countries, CVC use is even more common (Canada, 45%; Belgium, 38%; United Kingdom, 16%). Two types of CVC are used: nontunneled CVC for short-term use, and tunneled cuffed CVC, which can be used for several weeks to months or as permanent access in patients without AV access options. Nontunneled CVCs are associated with higher infectious complications and should be limited to less than 2 weeks duration of use. Box 96.1 provides indications for short- and long-term use of CVCs. Although the physical CVC characteristics (i.e., design and geometry) theoretically influence the performance (blood flow rate, recirculation, and resistance) and affect the overall efficiency of the HD therapy and morbidity risk

Nonautogenous Prosthetic Graft Vascular Access

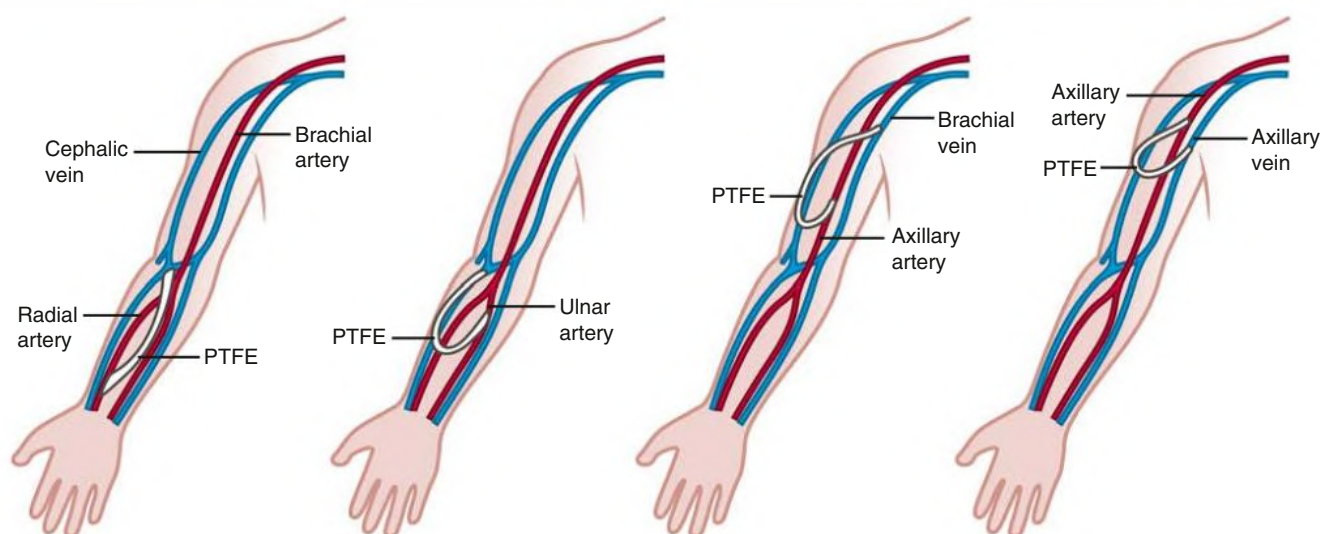


Fig. 96.4 Nonautogenous Prosthetic Graft (Polytetrafluoroethylene [PTFE]) Vascular Access. Straight and loop configuration in upper limb.

BOX 96.1 Indications for Short and Long-Term Use of Central Venous Catheters

- Temporary switch from another modality (PD-related complications; transplant acute rejection)
- Awaiting live-donor kidney transplant with established surgical date
- Short-term use of nontunneled CVC awaiting placement of tunneled CVC
- Temporary use if established AV access is temporarily unusable (e.g., maturing fistula); AV access requiring cannulation-free time due to complication (e.g., significant infiltration injury)
- Limited life expectancy (<6 months)
- Clinical conditions that worsen with AV access (heart failure with ejection fraction <15%; nontreatable skin lesions where cannulation increases infection or rupture risk)
- All other AV access options have been exhausted
- Patient choice after proper informed consent

AV, Arteriovenous; CVC, central venous catheter; PD, peritoneal dialysis.

(infection, thrombosis), no CVC type has been proven consistently superior to another.^{1,37}

Nontunneled Catheters

Single- or double-lumen CVCs are usually made of polymers (polyethylene, polyurethane), which enable easy insertion. The length of the CVC must be appropriately chosen to consider the anatomic insertion site. The femoral route requires catheters of 27 to 35 cm in length for the distal tip to be located in the inferior vena cava. The internal jugular vein route needs shorter catheters of 19 to 25 cm in length, with tip location at the mid-right atrium. The ideal CVC diameter is 12 to 14 Fr to allow sufficient blood flow to achieve adequate dialysis.

Tunneled Catheters

Tunneled CVCs have two lumens, each having a length of 40 cm, 10 cm of which is tunneled under the skin; the cannulae are made of synthetic

polymer with a large internal lumen and, typically, a Dacron cuff to ensure subcutaneous anchoring. The CVC characteristics rely on the type of polymer, design, and geometry (e.g., double-lumen CVC, dual CVC, split CVC). The use of tunneled CVCs is associated with reduced morbidity and consistently better performance compared with non-cuffed, nontunneled CVC.³⁸

Both tunneled and nontunneled CVC are inserted percutaneously by the Seldinger technique with ultrasound guidance and fluoroscopy. The internal jugular vein and femoral vein routes are preferred because of ease of insertion and lower risk of some complications, such as central vein stenosis, compared with subclavian vein insertion (Fig. 96.5). Radiologic placement of tunneled translumbar HD catheters (TLDC) is a safe and technically successful procedure in those with exhausted conventional CVC access. Compared with traditional CVCs, TLDCs have higher complication rates, exchanges, and removals (>30% of patients), usually due to poor blood flow and infection. These catheters function well in the short term but are typically considered a last resort.

VASCULAR ACCESS COMPLICATIONS

Primary Arteriovenous Fistula Problem: Nonmaturation

The AVF needs time to mature, allowing the vein to enlarge and toughen so that it can be needled for dialysis. Depending on the practice pattern and dialysis prescription, typically 6 to 12 weeks for maturation is required. AVF may be successfully cannulated by 2 weeks but can take as long as 9 months.^{35,39} Nonmaturation rates vary from 7% to 60%. The essential components of a successful AVF are a sufficient vein diameter of 4 to 5 mm for needling and Qa greater than 500 to 600 mL/min, for a typical dialysis prescription with a Qb of 300 to 400 mL/min, in order to prevent recirculation and achieve dialysis adequacy in a 4-hour time frame. Although these criteria have been used in research studies, clinical practice has informed that usability of AVF for cannulation is best judged by an experienced dialysis nurse with or without ultrasound vein diameter and access flow measurements. The statement of successful cannulators, such as “It’s the feel of the integrity and ‘toughness’ of the fistula, not just the size that determines

Tunneled Cuffed Double-Lumen Central Venous Catheter Inserted in the Right Internal Jugular Vein

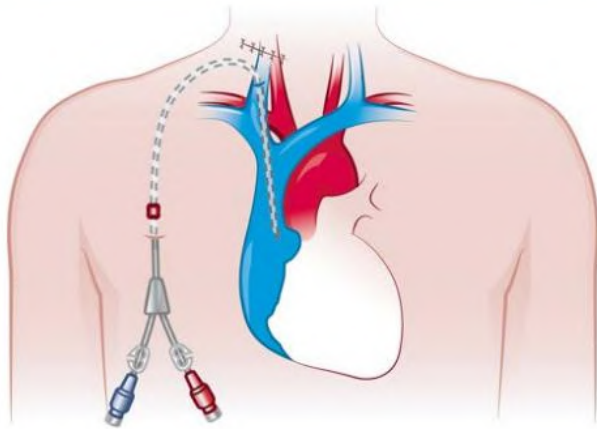


Fig. 96.5 Tunneled cuffed double-lumen central venous catheter inserted in the right internal jugular vein.

successful usability,” highlights that objective measurement should support, and not replace, cannulation experience.

Fistulas that fail immediately do so as a consequence of poor selection of vessels, poor surgical technique, or postoperative hemodynamic instability. New AVFs should be assessed by the operator within 2 weeks of creation with 4 to 6 weeks follow-up by the team if there is suspected nonmaturation. Almost all early, non-process-related AVF failures involve vascular abnormalities. Therefore, duplex ultrasound investigation can detect poor flow, stenosis, and accessory branches, guiding the interventional radiologist and surgeon to the appropriate treatment. More than half of the stenoses detected are in the perianastomotic area. Arterial inflow stenoses of more than 50% reduction in vessel diameter are seen in less than 10% of nonmaturing fistulas, but if identified, they should undergo angioplasty. Anastomotic and swing segment (in which the vein has been mobilized and swung over to the artery) stenosis may be treated percutaneously or surgically, depending on local expertise.

Percutaneous intervention includes angioplasty ($\pm$ drug coated balloons) of stenotic lesions, coil embolization, or ligation of accessory or collateral veins. In patients with residual kidney function (either predialysis or on HD), the risk of further loss of kidney function as a result of radiocontrast load can be mitigated by use of CO₂ angiography or by performance of ultrasound-guided angioplasty of the nonmatured fistula.^{1,40} Percutaneous transluminal balloon angioplasty (PTA) is further discussed in [Chapter 97](#).

When a surgical approach is selected, nonmatured AVFs should be reconstructed, usually under regional or local anesthesia. The anastomosis is exposed and ligated; the vein then can be divided, mobilized proximally, and reanastomosed to the proximal radial artery.

Limiting or preventing early failure is difficult, as the physiologic, functional, pathologic, and process of care factors that contribute to AVF failure or nonmaturation remain elusive.^{9,41,42} An RCT that compared brachial plexus block (BPB) with local anesthesia for AVF creation showed superiority of BPB, with significant improvement in the primary outcome of patency at 3 months. Consideration may be given to using BPB for AVF creation.⁴³ Hand and finger exercise may improve maturation and can be considered 2 weeks post AVF creation.¹ The use of far infrared therapy appears

promising, but its availability and generalizability is limited.^{1,44,45} To date, there are no consistent medical therapies found to improve maturation.

Primary Arteriovenous Graft Problem: Reduced Patency

Numerous studies have evaluated the influence of AV graft material and design on AV graft patency. Theoretically, modulating the distensibility (e.g., polyurethane grafts) or geometry of the arterial inlet or venous outlet of the AV graft may have a beneficial effect on intimal hyperplasia. However, clinical studies of polyurethane grafts and those using tapered (at the arterial side of the graft) AV grafts did not improve patency rates, nor did cuff implantation at the venous anastomosis. Cuff-shaped prosthesis (e.g., Venaflow) may be beneficial.^{46,47}

Because ePTFE AVGs are prone to thrombosis, infection, and intimal hyperplasia at the venous anastomosis, tissue-engineered vascular grafts have been developed as a potential solution to these limitations. Preliminary results of human studies suggest these biologic grafts are safe and functional; further phase III studies are underway.⁴⁸

Both Aggrenox and fish oil have been found to improve primary patency of AV grafts.^{49,50} Fish oil has also been shown to reduce the rate of thrombosis and interventions. Given its low cost and safety, fish oil can be considered in patients who need AV graft access.¹

General Arteriovenous Access Problem: Stenosis and Thrombosis

Vessel stenosis is common in both autogenous AVFs and prosthetic AVGs. Intimal hyperplasia, with smooth muscle cell migration, proliferation, and matrix deposition, is the major cause of stenosis formation and thrombosis. The cause of the intimal hyperplasia is multifactorial, with uncertain contributing factors. They range from the uremic environment, the impact of surgery, the high wall shear stress caused by the access flow itself, and the downstream effect of the denuded endothelial cell layer, resulting in platelet adhesion and initiation of a cascade of proteins that stimulate the smooth muscle cells to proliferate and to migrate.^{51,52} Progressive stenosis leads to access flow deterioration and subsequently thrombotic occlusion. Prophylactic repair of access stenoses may prevent thrombosis and prolong access patency.

Autogenous Arteriovenous Fistula Stenosis or Thrombosis

AVF stenosis should be suspected if the vessel diameter is reduced by more than 50% and is accompanied by clinical indicators such as difficulties with cannulation or prolonged bleeding time after decannulation (indicating high intraaccess pressure caused by outflow vein stenosis). [Box 96.2](#) discusses clinical indicators. In AVFs, 55% to 75% of the stenoses are close to the AV anastomosis, 25% in the venous outflow tract, and 15% in the arterial inflow. In brachiocephalic and brachio basilic AVFs, stenosis is typically located (other than the anastomosis) at the junction of the cephalic with the subclavian vein (cephalic arch) and the basilic with the axillary vein (junctional stenosis). Initial evaluation of suspected access stenosis is by ultrasound; angiography is used subsequently when intervention is planned.

Endovascular treatment by PTA is the first option for arterial inflow and venous outflow stenoses and junctional stenoses, with the option of stent or stent-graft placement.⁵³ These techniques are discussed further in [Chapter 97](#). Use of stents or stent grafts should avoid cannulation segments.¹ Some stenoses may not be sufficiently dilated by conventional balloons (12–16 atm), and in these accesses, cutting balloons or ultra-high-pressure balloons (up to 36 atm) may be applied. Drug-eluting balloons have shown to be beneficial in preventing restenosis; however, data are conflicting.^{54,55} Surgical revision may be appropriate, depending on the nature and location of the lesion; for

BOX 96.2 Clinical Indicators of Vascular Access Stenosis

Physical Examination

- Ipsilateral extremity edema
- Alterations in pulse, with a weak or resistant pulse, difficult to compress in the area of the stenosis
- Abnormal thrill (weak and/or discontinuous) with only a systolic component in the region of the stenosis
- Abnormal bruit (high pitched with a systolic component in the area of stenosis)
- Failure of the fistula to collapse when the arm is elevated (outflow stenosis) and lack of pulse augmentation (inflow stenosis)
- Excessive collapse of the venous segment upon arm elevation

Dialysis

- New difficulty with cannulation when previously not a problem
- Aspiration of clots
- Inability to achieve the target dialysis blood flow without any other cause
- Prolonged bleeding beyond usual for that patient from the needle puncture sites for three consecutive dialysis sessions
- Unexplained decrease in the delivered dialysis dose (Kt/V) on a constant dialysis prescription without prolongation of dialysis duration, and other variables excluded

example, a more proximal reanastomosis for swing segment stenosis is indicated after failed PTA of a radiocephalic AVF.

Fistula thrombosis should be treated as soon as possible because timely declotting allows immediate use of the AVF without the need for a CVC. Fistula intervention is ideal within 6 hours, and success is greatly reduced after 72 hours (grafts should be intervened on within 72 hours but can be salvaged up to 5–7 days). The duration and site of AV access thrombosis, as well as the type of AV access, are important determinants of treatment outcome. Thrombi become progressively fixed to the vein wall, which makes surgical removal more difficult. When the clot is localized at the anastomosis in radiocephalic and brachiocephalic AVFs, the outflow vein may remain patent because of continuing flow in its tributaries, making it possible to create a new proximal anastomosis.⁵⁶

Thrombolysis can be performed mechanically or pharmacomechanically. Whereas the immediate success rate is higher in AVGs than in AVFs (99% vs. 93% in forearm AVFs), the primary patency rate of the forearm AVF at 1 year is much higher (49% vs. 14%). One-year secondary patency rates are 80% in forearm and 50% in upper arm AVFs.⁵⁷

Nonautogenous Arteriovenous Graft Stenosis or Thrombosis

The most common cause of AV graft dysfunction and thrombosis is venous anastomotic stenosis. Because AV grafts are typically created in patients with exhausted peripheral veins, vein-saving procedures such as PTA or patch angioplasty are preferred to AV graft extensions to more central venous segments. When a stent or a patch fails, AV graft extension is still possible. Clinical monitoring of an AV graft is recommended (see Box 96.2) and can be supplemented with surveillance, such as access flow (Qa) measurement[†]; ultrasound detection of stenoses with preemptive endovascular treatment may diminish AV graft thrombosis but does not extend AV graft patency.

Intragraft (or midgraft) stenoses are found in the cannulation segment of AV grafts. They result from excessive ingrowth of fibrous tissue through puncture holes. These stenoses can be treated by PTA, AV graft curettage, or segmental AV graft replacement. When only a part

of the cannulation segment is replaced, the access can be used immediately for HD without the need for a CVC. When restenosis occurs in a nonexchanged part of the AV graft, this can be replaced after healing of the new segment.

AV graft thrombosis can be treated with various percutaneous techniques and tools, including combinations of thromboaspiration, thrombolytic agents such as urokinase, tissue plasminogen activator (tPA), and mechanical thrombectomy. An initial success rate of 73% and primary patency rates of 32% and 26% at 1 and 3 months, respectively, are reported.⁵⁷ It is ideal to perform thrombolysis as soon as possible to avoid CVC use and as an outpatient procedure for best clinical outcomes and cost containment. Postprocedural angiography to detect and correct inflow, intraaccess, or venous outflow stenosis is mandatory.

When endovascular treatment fails or is not possible, surgical thrombectomy may be performed with a Fogarty catheter after venotomy, with correction of the underlying obstruction. On-table angiography should be performed after completion of thrombectomy of both the arterial and venous limbs of the graft.

Central Venous Obstruction

In the majority of patients, central vein obstruction is a result of previously inserted CVCs or pacemaker wires. In 40% of patients with subclavian vein catheters and in at least 20% with jugular vein catheters, central venous stenosis or occlusion will develop. Chronic swelling of the access arm is the most obvious sign, usually with prominent superficial collateral veins around the shoulder. The indications for intervention by PTA and stent placement are severe and disabling arm swelling, pain, finger ulceration, and inadequate HD. Contrast angiography of the access and complete venous outflow tract must be performed; ultrasound is not suitable for examining the central veins because of their retroclavicular location.

The benefits and risks of AVF, AVG, and CVCs are listed in Table 96.4.

Interventions to Manage Patency Complications

Endovascular intervention. Endovascular intervention is the first treatment option for central venous obstruction. PTA alone results in patency rates of only 10% or less at 1 year, and numerous restenoses may develop. Primary or additional stent implantation is associated with a better outcome, with 1-year patency rates up to 56% or higher.⁵⁸ Reinterventions are usually required to maintain patency and achieve long-term clinical success.

Stent placement should avoid overlapping the ostium of the internal jugular vein because this vein is essential for future placement of CVCs. Similarly, a stent placed in the innominate vein should not overlap the ostium of the contralateral vein; otherwise, contralateral stenosis may occur and preclude future creation of a contralateral AV access.

A percutaneous approach to bypass central venous obstruction with the “inside out” technique using the Surfacter System Inside-Out Access Catheter System is a novel and daring method for gaining central venous access.⁵⁹

Surgical intervention. When endovascular treatment of central venous obstruction fails, surgical revision with bypass grafting is indicated. Surgical bypass to the ipsilateral jugular vein or contralateral subclavian or jugular vein is the first option in these patients. Alternative surgical approaches for upper limb vascular accesses with compromised venous outflow are axillary vein to femoral, saphenous, or popliteal vein and right atrial bypasses.⁶⁰ In case of bilateral obstruction of the mediastinal veins, including the superior vena cava, it will not be possible to sustain upper limb access, and lower limb access will be required.

TABLE 96.4 General Benefits and Risks of AV Fistula, AV Graft, and CVC

AV Fistula	AV Graft	CVC
high maturation failure	low maturation failure	no maturation needed
difficult cannulation	easy cannulation	easy connection to dialysis machine
low-moderate infection rate	low-moderate infection rate	moderate-high infection rate
low-moderate revision rate	moderate-high revision rate	high revision rate
good long-term patency if maturation with few facilitative interventions	moderate long-term patency	short long-term patency
low thrombosis rate	moderate thrombosis rate	high thrombosis rate

AV, Arteriovenous; CVC, central venous catheter.

Ultimately, ligation of the upper limb access can be considered, which will relieve local symptoms but sacrifices a valuable dialysis access.

Pharmacologic approaches for access patency. Based on clinical trials, there is insufficient evidence to determine whether medical treatments such as ticlopidine, aspirin, fish oil, clopidogrel, dipyridamole, warfarin, and sulfinpyrazone improve the patency of AV accesses compared with placebo treatment; the quality of the evidence is low because of short follow-up periods, the small number of studies for each comparison, heterogeneity among trials, and incomplete reporting.⁶¹ Warfarin may reduce AVG thrombosis but increases the risk for hemorrhage and vascular calcification.⁶² A large trial showed that dipyridamole plus aspirin had a statistically significant but modest clinical effect in prolonging the duration of primary unassisted patency of newly created AVGs.⁴⁹ Fish oil reduced AVG thrombosis and also improved unassisted patency of newly created AVG in RCTs.^{63,64} In a large randomized study, clopidogrel improved fistula thrombosis but not functional maturation and usability.⁶⁵

Clinical trials in patients undergoing AVF and AVG creation, who had perivascular biodegradable collagen wraps with sirolimus placed at the anastomosis, demonstrated safety and feasibility of the sirolimus wrap. A phase III RCT is ongoing to assess the safety, efficacy, and patency outcomes of a perivascular sirolimus-eluting implant placed at the AVF anastomosis in HD patients receiving new AVFs.⁶⁶

Vascular Access–Induced Ischemia

Vascular access–induced upper limb ischemia is a serious complication that without prompt intervention may lead to amputation. The incidence of symptomatic ischemia varies from 2% to 8% of the HD population.⁶⁷ Elderly patients, diabetics, patients with prior ipsilateral AV access, and those with peripheral or coronary arterial occlusive disease are most at risk for ischemia. Access-induced ischemia occurs more often with proximally located AVFs and high flow rates; such AVFs can induce a steal phenomenon with lowering of distal perfusion pressures, and when collateral circulation is inadequate, symptoms may occur. Pain during HD is a characteristic early symptom. A grade 1 to 4 classification for access-induced ischemia can be used to outline the severity of the disease; this ranges from minor symptoms to finger necrosis.⁶⁸

- Grade 1: Pale and blue or cold hand without pain
- Grade 2: Pain during exercise or HD
- Grade 3: Ischemic pain at rest
- Grade 4: Ulceration, necrosis, and gangrene

For grades 1 and 2, conservative treatment of ischemia is advocated. With grades 3 and 4, interventional treatment is mandatory.

Diagnosis of Ischemia

Physical examination, including observation and palpation of peripheral vessels, may be inadequate and misleading for the diagnosis of

symptomatic ischemia. Additional noninvasive testing with measurement of digital pressures and calculation of the digit-brachial index, transcutaneous oximetry (TcPo₂), ultrasound of forearm arteries, and access blood flow measurement are important steps in the diagnosis and decision-making process.⁶⁹ Finally, contrast angiography with visualization of the upper extremity arterial tree from the proximal subclavian artery to the distal palmar arches with and without AVF compression to enhance distal flow is obligatory to outline the strategy for treatment and determine whether endovascular or surgical options are preferred.

Endovascular and Surgical Management of Ischemia

The treatment strategy depends on the cause of the ischemia. Inflow arterial obstruction and distal arterial lesions are recanalized with small-caliber balloons or stent placement⁷⁰; high-flow AVFs are suitable for flow-reducing procedures such as access banding and arterial inflow reduction by an interposition graft to a smaller forearm artery (revision using distal inflow [RUDI]) (Fig. 96.6).^{71,72} Steal in itself may be corrected by ligation of the artery distal to the AV anastomosis or revascularization procedures such as the distal revascularization–interval ligation (DRIL) procedure (Fig. 96.7) or other procedures.⁷³

Intraoperative digital pressure measurement (or TcPo₂) is mandatory to guarantee an adequate surgical intervention with acceptable outcome. A digital-brachial pressure index above 0.60 or TcPo₂ above 40 mm Hg is indicative of sufficient distal hand perfusion. In some patients, AVF ligation and transition to chronic CVC access or a change of dialysis modality to peritoneal dialysis may be the only solution.

Catheter-Related Infection

Catheter-related bloodstream infections are a significant cause of morbidity and mortality in HD patients. CVC use is associated with a higher risk of mortality compared with AV access.¹ Switching from CVCs to an AV access is associated with a lower risk of mortality compared with continuing CVC use.⁷⁴ CVC-related infections include bacteremia, tunnel infections and exit site infections. Any of these types of infection can progress to metastatic complications of osteomyelitis, septic arthritis, epidural abscess, and endocarditis. Nontunneled CVC have a higher rate of infection compared with tunneled CVC.³⁸ Each facility should track and monitor CVC-related infection rates, with a goal bacteremia rate of less than 1.5 per 1000 CVC days.¹

Prevention of Catheter-Related Infection

The most important measure to prevent CVC infection is meticulous handling of the CVC at all times. The CVC should be inserted with use of maximal infection control precautions, per local facility guidelines. The dialysis nurses, technicians, or patient/caregiver (if home dialysis) needs to access the CVC under strict aseptic or sterile conditions; it is of the utmost importance that CVCs are never accessed by untrained

Surgical Techniques for Banding of a High-Flow Vascular Access

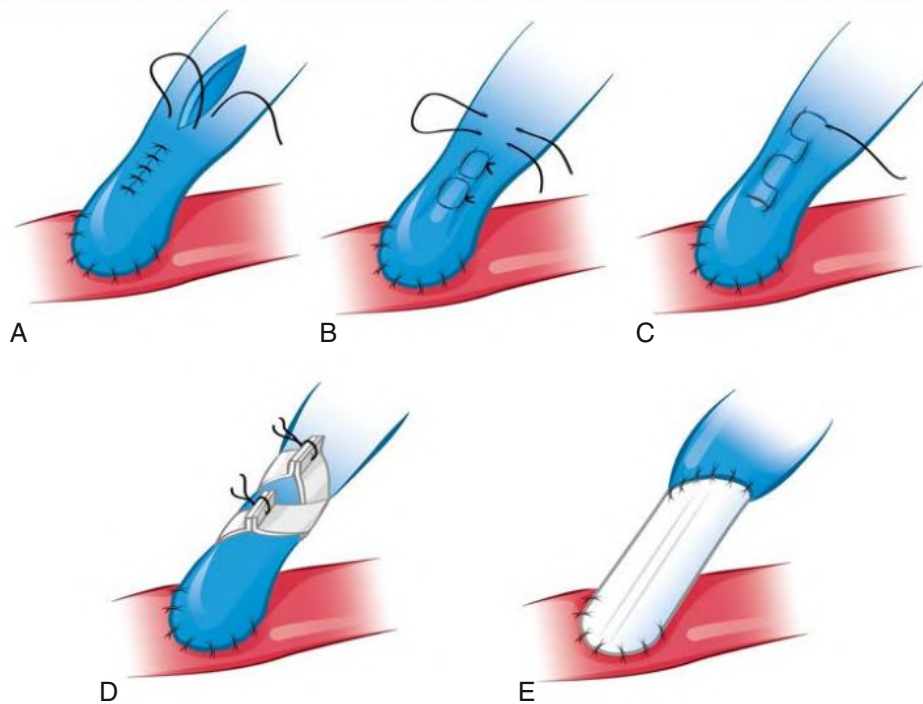


Fig. 96.6 Surgical Techniques for Banding of a High-Flow Vascular Access. (A) Open venoplasty. (B) Interrupted mattress suturing. (C) Continuous mattress suturing. (D) Polytetrafluoroethylene (PTFE) banding. (E) PTFE interposition graft. The choice of technique is made by the surgeon on a case-by-case basis.

personnel. An exemplar protocol for safe connection and disconnection of CVCs can be found in the KDOQI Vascular Access Guidelines 2019.¹ Catheter locking with trisodium citrate solutions appears to reduce bleeding and may reduce bacteremia compared with locking with heparin solutions.⁷⁵ However, it is unclear whether citrate alone has the same protective effect against systemic infection as it does when combined with other antimicrobial agents^{76,77} and may depend on the concentration of citrate used.⁷⁷ Mupirocin ointment applied to the exit site may reduce the incidence of CVC-associated bacteremia, but there are concerns of CVC material breakdown and mupirocin resistance. Topical application of povidone, medi-honey, and Polysporin triple ointment to exit sites has been shown in multicenter RCTs to reduce the rate of CVC-related bacteremia. Polysporin triple has sustained efficacy with no demonstrated resistance after long-term follow-up.^{1,78}

Treatment of Catheter-Related Infection

Managing catheter-related infections involves treating the patient and managing the catheter. Patients should have appropriate culture and sensitivities obtained and treated with antibiotics according to the results of the sensitivities. Until they return, the infected patient should be treated with empiric broad-spectrum antibiotics to cover both gram-positive and gram-negative organisms.

Various societies have issued recommendations for the management of catheter infections.^{1,79} A recommended treatment algorithm is shown in Fig. 96.8. A more detailed algorithm can be found in the KDOQI Vascular Access Guidelines 2019, including durations of antibiotic use and special management considerations depending on the infecting organism. For example, if the infecting organism is *Staphylococcus aureus* causing bacteremia, an infectious disease consult may be considered. All such patients should undergo a 2D echocardiogram to evaluate for the presence of vegetations or

endocarditis, and antibiotic treatment should be at least 4 weeks in duration.

Infections Involving Temporary Nontunneled Catheters

When a temporary nontunneled CVC becomes infected, it should always be removed. There is no role for trying to salvage temporary nontunneled catheters.¹

Exit Site Versus Tunnel Tract Infections

An exit site infection is a localized cellulitis confined to the 1 to 2 cm where the catheter exits the skin. The majority of these infections respond well to oral antibiotics and meticulous exit site care, and the removal of the CVC is generally not required.^{1,78} However, exit site infections can progress to tunnel tract infections, which involve the potential space surrounding the CVC more than 2 cm from the exit site (Fig. 96.9). Tunnel infections should be treated with intravenous antibiotics. Patients with a tunnel infection sometimes but not always have an associated exit site infection; untreated, they can rapidly develop bacteremia. Patients with a tunnel infection present with fever and local signs of pain, swelling, fluctuance, and erythema along the tract of the catheter. There may be purulent discharge from the exit site. Because tunnel infections involve a potential space in an area with limited vascular supply and an implanted synthetic device, they respond poorly to antibiotics alone and require catheter removal.^{1,78}

Catheter-Associated Bacteremia

When a patient with a dialysis catheter has a fever, CVC-related bacteremia must be considered but other causes ruled out. If the patient does not have a clear and convincing alternative explanation for the fever, blood cultures should be obtained peripherally (e.g., dialysis circuit), as well as through the catheter hub⁸⁰; as indicated earlier, the

Distal Revascularization-Interval Ligation (DRIL) for Ischemia in an Upper Arm Vascular Access

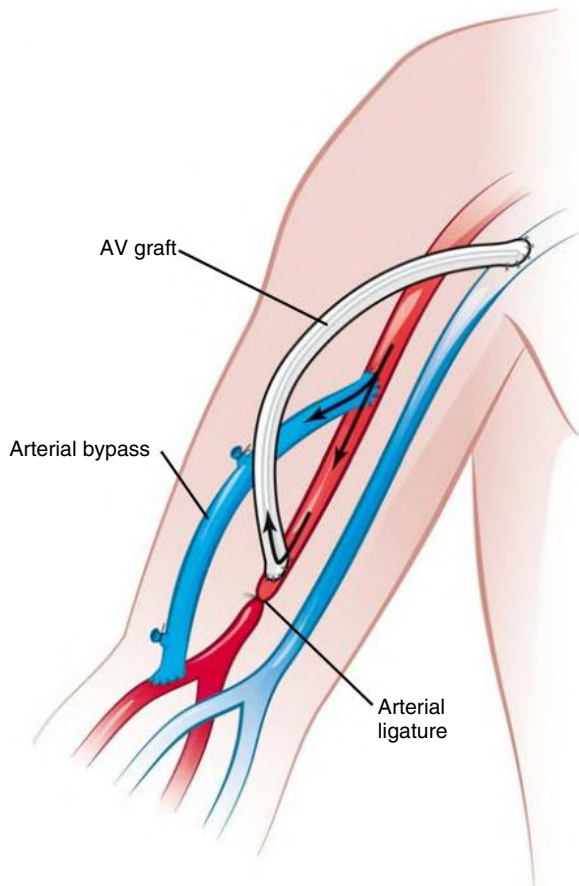


Fig. 96.7 Distal revascularization–interval ligation for ischemia in an upper arm vascular access. AV, Arteriovenous.

patient should be started on empiric antibiotics and adjusted based on culture results.¹ The most common organism is *S. aureus*, although a wide range of gram-positive and gram-negative organisms have been reported. An aminoglycoside or a cephalosporin is a good choice for gram-negative coverage; however, the presence of residual kidney function and local microbiologic epidemiology must be considered, especially with regard to antibiotic resistance.

Management of Catheter-Related Infection

The decision to remove a tunneled cuffed CVC because of an episode of CVC-associated bacteremia is not straightforward. The clinical condition of the patient and response to initial therapy, the presence of an infected fibrin sheath, metastatic complications, the infecting organism, and the availability of other vascular access sites must be taken into consideration (see Fig. 96.8).

The conventional approach is to remove the CVC with interval replacement at a different site after the infection has resolved. The efficacy of this approach is uncertain, and it requires temporary CVC insertion before a new CVC is placed. Further, if the CVC is removed without identifying and disrupting an infected fibrin sheath, a nidus of continual infections remains. A new CVC may inadvertently be placed back into the infected sheath. The ineffectively treated nidus of infection may explain why attempts to salvage an infected CVC with systemic antibiotic therapy is only successful in about 30% of patients. Another treatment option is to combine systemic antibiotics with antibiotic “lock” solutions. Many different combinations of antibiotics mixed with either heparin or citrate have been tested; the appropriate antibiotic lock depends on the culture sensitivities. A suitable broad-spectrum regimen is vancomycin 2.5 mg/mL, gentamicin 1 mg/mL, and heparin 2500 U/mL. Infection clearance rates of 50% to 70% are reported with antibiotic locking.

CVC exchange over a guidewire after 48 hours of antibiotic treatment can be more effective than treatment with antibiotics alone and is at least as effective as removal of the CVC and delayed replacement, with the advantages of only one invasive procedure and preservation of the venous access site.^{76,81} This also has the advantage of disrupting a fibrin sheath, if present.^{82,83}

Management of Central Venous Dialysis Catheter Infections

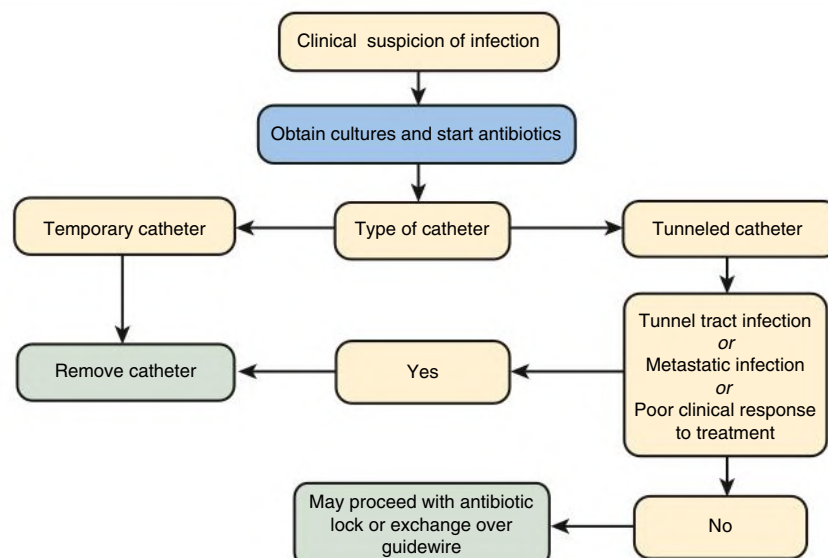


Fig. 96.8 Algorithm for the management of central venous dialysis catheter infections. (Modified from Manian FA. IDSA guidelines for the diagnosis and management of intravascular catheter-related bloodstream infection. *Clin Infect Dis.* 2009;49[11]:1770–1772.)



Fig. 96.9 Dialysis catheter tunnel infection. (Courtesy Dr. I.M. Leidig, University Hospital Erlangen, Germany.)

Catheter Thrombotic Obstruction

Catheter obstruction may be caused by endoluminal fibrin deposits, restricting the CVC lumen or obstructing CVC side holes or at the tip,

or external fibrin sleeves surrounding the CVC, resulting in inadequate flow and excessive extracorporeal blood pressure alarms during the dialysis session. Depending on the location of the fibrin clot (arterial or venous line), there may be high negative arterial pressure (obstruction at the arterial CVC) or high positive venous pressure (obstruction at the venous CVC).

Prevention of clot formation in the CVC lumen and tip during the interdialytic period is crucial. This is achieved by installing an anticoagulant lock solution (heparin or trisodium citrate).¹ Catheter malfunction might be treated with regular thrombolytic locking (e.g., urokinase or tPA); further, routine prophylactic locking was found to reduce CVC malfunction and CVC-related bacteremia in a multicenter RCT.⁸⁴ RCTs have shown that regular use of low-dose warfarin or antiplatelet agents does not improve CVC function but increases the risk of major bleeding in HD patients.^{1,62}

There are few established maneuvers to prevent CVC-related malfunction beyond the use of regular tPA, as demonstrated in the PRECLOT study⁸⁴; however, the latter may become cost prohibitive. If a patient depends on CVC for HD access, the CVC may require multiple exchanges over guidewires. It is therefore critical to properly lay out the patient's ESKD Life-Plan and have a PLAN for each patient in order to appropriately limit CVC use and maximize safe, complication-free vascular access options for each patient.

SELF-ASSESSMENT QUESTIONS

- The PLAN includes which of the following components?
 - The patient's ESKD Life-Plan only
 - A plan for the patient's individualized vascular access care only
 - The patient's ESKD Life-Plan and corresponding access needs (insertion/creation plan, contingency plan, succession plan, vessel preservation plan)
 - The plan for the hemodialysis access (insertion/creation plan, contingency plan, succession plan, vessel preservation plan)
- AV access (fistulas and grafts) is preferred to CVCs because of:
 - the lower rate of thrombotic complications.
 - the lower rate of infections.
 - the lower rate of nonthrombotic complications.
 - all the above.
- Ischemia in elbow fistulas with adequate access flow (800 mL/min) is best treated by:
 - access banding.
 - the DRIL procedure.
 - the revision using distal inflow procedure.
 - distal radial artery ligation.
- Tunneled cuffed CVCs are preferred over nontunneled catheters because of:
 - ease of insertion.
 - lower occlusion rate.
 - higher dialysis flows.
 - lower infection rate.
- The primary treatment option for symptomatic central vein occlusion is:
 - percutaneous transluminal angioplasty.
 - stent placement.
 - surgical bypass.
 - vascular access ligation.

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